

Scaling Book Exercises: Section 12, Quiz 5

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Quiz: LLM Rooflines

Source scan: ../handwritten/section_12_handwritten.pdf, pages 11–13

Reference: <https://jax-ml.github.io/scaling-book/gpus/>

Question 1: B200 rooflines

A B200 provides 2250 bf16 TFLOPs/s versus 990 for H100, a $2.25\times$ increase. Within an eight-GPU node, egress bandwidth doubles from 450 to 900 GB/s. The B200 critical ratio is therefore

$$\frac{2250 \times 10^{12}}{900 \times 10^9} = 2500.$$

For tensor parallelism the approximate limit is $Y < F/2500$, close to the H100 limit because compute and within-node communication improved together.

Beyond one node, egress remains 400 GB/s while compute increases. The data-parallel critical per-GPU token batch becomes

$$\frac{2250 \times 10^{12}}{400 \times 10^9} = 5625 \text{ tokens/GPU.}$$

Thus B200 improves compute faster than scale-out bandwidth and makes the cross-node communication roofline harder to satisfy.

Question 2: Sharding LLaMA 3-70B

(a) Storage lower bound

Training in bf16 with Adam requires approximately two bytes per parameter for weights and eight bytes for fp32 optimizer state:

$$70 \times 10^9(2 + 8) = \boxed{700 \text{ GB.}}$$

With 80 GB per H100, the raw lower bound is

$$\left\lceil \frac{700}{80} \right\rceil = \boxed{9 \text{ GPUs}},$$

or two eight-GPU nodes after rounding to the node topology. This excludes gradients and activation checkpoints.

(b) Training time

The standard transformer estimate is six FLOPs per parameter per training token:

$$\text{total FLOPs} = 6(70 \times 10^9)(15 \times 10^{12}) = 6.3 \times 10^{24}.$$

At 45% MFU on 4096 H100s, effective throughput is

$$4096(990 \times 10^{12})(0.45) \approx 1.825 \times 10^{18} \text{ FLOPs/s}.$$

Therefore

$$T \approx \frac{6.3 \times 10^{24}}{1.825 \times 10^{18}} = 3.45 \times 10^6 \text{ s} \approx 40 \text{ days}.$$

(c) Parallelism and communication

For $F = 28,672$, the within-node H100 model-parallel roofline suggests

$$\frac{F}{1995} \approx 14.4.$$

Because an H100 node contains eight GPUs, the practical choice is $\boxed{8}$ -way tensor parallelism.

Pure data parallelism would then require $4096/8 = 512$ replicas, but each GPU would hold

$$\frac{700 \text{ GB}}{8} = 87.5 \text{ GB},$$

which exceeds its 80 GB HBM.

With ZeRO-3 plus eight-way tensor parallelism, memory fits, but the local batch is only

$$\frac{4 \times 10^6}{4096} \approx 976 \text{ tokens/GPU}.$$

This is below the data-parallel roofline, and ZeRO-3 also communicates parameters, so it remains communication-bound.

Adding eight-way pipeline parallelism makes each model shard span eight nodes. This raises the effective scale-out bandwidth to

$$8(400) = 3200 \text{ GB/s},$$

giving a critical batch of

$$\frac{990 \times 10^{12}}{3200 \times 10^9} \approx 309 \text{ tokens/GPU}.$$

The 976-token local batch is above this bound, so efficient pipelining can make the configuration compute-bound.

Question 3: Megatron-LM configurations

With sequence length 4096, the per-GPU token batches are

$$\begin{aligned} 16\text{B: } \frac{192(4096)}{192} &= \boxed{4096}, \\ 70\text{B: } \frac{384(4096)}{768} &= \boxed{2048}, \\ 314\text{B: } \frac{1536(4096)}{3072} &= \boxed{2048}. \end{aligned}$$

The unadjusted spine-level threshold is approximately 2472 tokens/GPU. The 16B model is already above it. The 70B and 314B models appear below it until their model parallelism is included:

Model	Model parallelism	Adjusted threshold	Result
16B	—	2472	compute-bound
70B	16×	$\approx 2472/2 = 1236$	compute-bound
314B	64×	$\approx 2472/8 = 309$	compute-bound

The 16×

 and 64× model-parallel layouts span approximately two and eight nodes, improving effective spine throughput by 2× and 8×, respectively. Hence all three configurations are theoretically compute-bound under this roofline model.